



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

Semester II 2023/2024

Subject : **SECD2613 System Analysis and Design**
Task : **Assignment 1**
Title : **Group Assignment 1 (Project Planning 5%)**
Due : 19/4/2024 5:00 PM
Submission : Submit softcopy (in .pdf format) via e-learning

Group Name: Systech		
Full Name	Matric Number	Designation
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Question 1

a. Assessing the feasibility of the systems project Ahmad is proposing requires considering various factors.

Technological feasibility involves evaluating whether the proposed changes are technically achievable given the current state of technology. In this case, Abu, as the systems analyst, would need to assess whether the existing technology can support the proposed system upgrades and whether implementing them would result in technical challenges or limitations. Additionally, considerations such as compatibility with existing systems and potential disruptions to operations need to be taken into account.

b. Based on Abu's interactions with managers, users, and systems people, the operational feasibility of the proposed project appears to be relatively low. Managers and users express satisfaction with the current system, indicating minimal perceived need for change. Additionally, Abu acknowledges the potential disruption that a new system could cause, highlighting concerns about employee resistance to change. While the technology to update the system is available, Abu's assessment suggests that the organization may not be ready for such a significant overhaul. Therefore, despite the technical feasibility, the operational feasibility of the proposed project is limited by the lack of perceived necessity and potential resistance from stakeholders.

c. Economic feasibility involves evaluating the cost-effectiveness of the proposed project. Ahmad mentions concerns about the significant financial investment required for implementing a new system. Abu also raises the point that the current system is functioning adequately, suggesting that the potential benefits of upgrading may not justify the associated costs. Therefore, from an economic perspective, the feasibility of the proposed project appears questionable, especially considering the uncertainty surrounding the return on investment. The economic feasibility of updating AllFoods' system hinges on conducting a thorough Cost-Benefit Analysis (CBA) to determine whether the time and financial resources required for development align with the potential benefits. This entails assessing tangible and intangible benefits, such as increased efficiency and improved reputation, against the costs of new equipment, hardware, software, and ongoing operational expenses. Techniques like break-even analysis, cash-flow analysis, payback period evaluation, and present value analysis can help quantify these factors and inform the decision-making process. Ultimately, the economic feasibility rests on whether the projected benefits outweigh the investment required, ensuring that the company's resources are utilized effectively to drive sustainable growth and competitiveness.

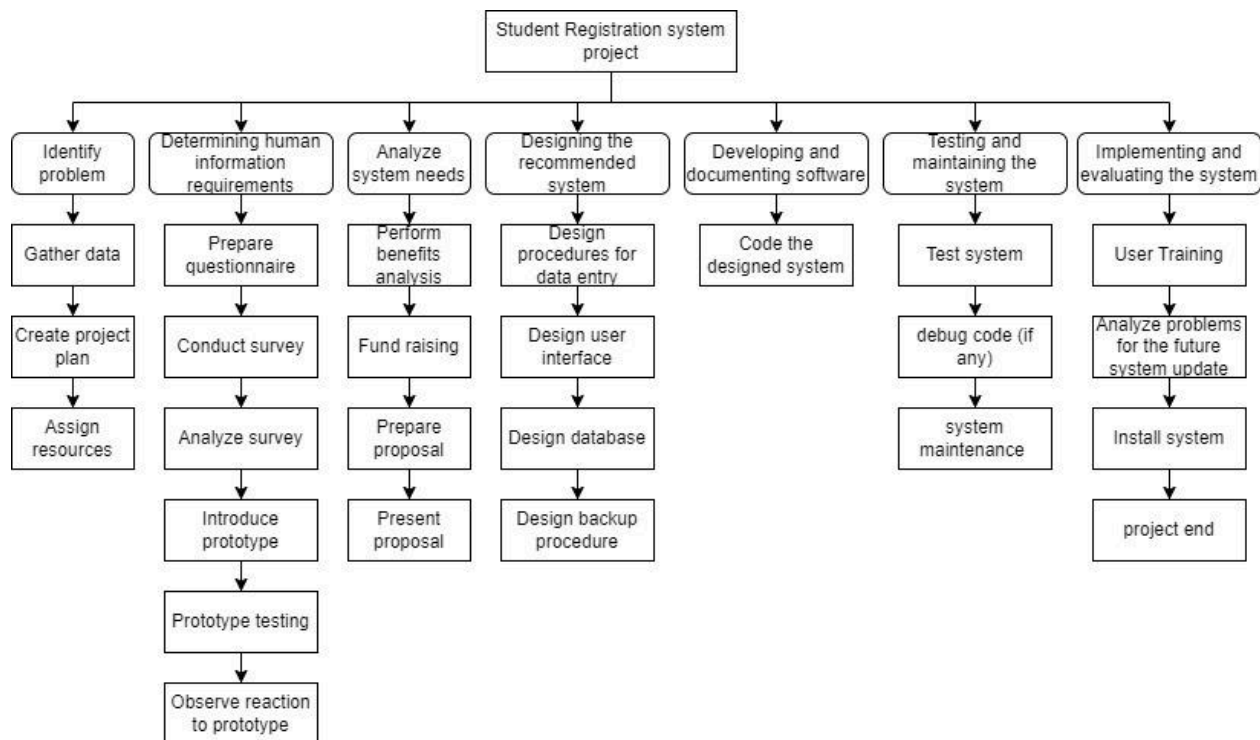
d. Technological feasibility refers to whether the proposed changes can be implemented using available technology. Abu confirms that the technology to update the system is indeed available. However, the presence of technology does not necessarily guarantee feasibility. The feasibility also depends on factors such as compatibility, scalability, and potential integration challenges with existing systems. In assessing the technological feasibility of updating AllFoods' system, several factors need consideration. Firstly, it's crucial to determine whether the current technical resources are sufficient for implementing the new system. Abu's interviews with staff members indicate satisfaction with the existing system, suggesting that it may be technically sound. However, if upgrades are needed to meet the technological requirements of the proposed changes, the feasibility hinges on whether these upgrades are feasible and cost-effective. Additionally, Abu's conversations with the systems team confirm that the technology to update the system is available, but his response suggests that upgrading solely for the sake of adopting new technology may not be advisable. Thus, while the technical resources for updating the system may be present, the feasibility also depends on whether such changes align with the organization's needs and strategic goals, as well as whether the potential disruptions are justified by the benefits.

e. Given the discussions between Ahmad and Abu, it seems prudent to recommend conducting a full-blown systems study before proceeding further. While Ahmad emphasizes the potential benefits of modernizing the system to improve the company's image, Abu raises valid concerns about the operational, economic, and technological feasibility of the proposed project. A comprehensive systems study would provide a detailed assessment of these factors, allowing the company to make a well-informed decision regarding whether to proceed with the proposed changes. This study would involve analyzing current system capabilities, potential impacts on operations, cost-benefit analysis, and technological requirements. Ultimately, a thorough assessment would mitigate risks and ensure that any proposed changes align with the company's goals and resources.

Question 2

WBS of new computer system development for student registration

Task ID	Tasks	Predecessor Tasks	Estimated Duration (days)
1	Identify problem	-	-
1.1	Gather data	None	3
1.2	Create project plan	1.1	3
1.3	Assign resources	1.2	1
2	Determining human information requirements	-	-
2.1	Prepare questionnaires	1.3	1
2.2	Conduct survey	2.1	7
2.3	Analyse survey	2.2	2
2.4	Introduce prototype	2.3	7
2.5	Prototype testing	2.4	5
2.6	Observe reaction to prototype	2.5	3
3	Analyse system needs	-	-
3.1	Perform benefits analysis	2.5	2
3.2	Fund raising	3.1	7
3.3	Prepare proposal	3.2	7
3.4	Present proposal	3.3	1
4	Designing the recommended system	-	-
4.1	Design procedures for data entry	3.4	5
4.2	Design user interface	4.1	5
4.3	Design database	4.2	5
4.4	Design backup procedure	4.3	5
5	Developing and documenting software	-	-
5.1	Code the system	4.4	20
6	Testing and maintaining the system	-	-
6.1	Test system	5.1	3
6.2	Debug code	6.1	3
6.3	System maintenance	6.2	10
7	Implementing and evaluating the system	-	-
7.1	User training	6.3	3
7.2	Analyse problems for the future system update	7.1	5
7.3	Install system	7.1	2
	Project end		



Question 3

Nowadays, many projects involve team members that are dispersed across various time zones and locations. Moreover, the adoption of agile methodology has improved flexibility and efficiency in project management. This report highlights key differences in managing projects in distributed agile teams compared to traditional project settings.

First of all, the traditional method provides clear steps to follow to achieve a goal and the project manager must approve any change, while agile project management allows for priorities to shift throughout a project. Besides that, traditional project management might include weekly meetings with a team and the project manager may meet with other stakeholders, while an agile team usually will meet every day for a short time to identify any issues, the project status is also typically visible to all stakeholders. The third difference is about escalation. In traditional project management, the project manager involves the team, evaluates impact and makes the final decision when resolving an issue, while in an agile environment, the entire team collaborates and agrees on a resolution. Next, whether it is software or a new process, dedicated teams perform testing near the end of a project in the traditional method, while an agile team includes users and customers to test regularly with each iteration.

In conclusion, managing agile projects requires a shift in mindset. It involves embracing flexibility, collaboration, and responsiveness. Meanwhile the traditional project management emphasizes predictability and adherence to a predefined plan.

Question 4

(a)

Costs	Year 0	Year 1	Year 2	Year 3
Development Costs				
- Hardware	50 000			
- Software	15 000			
- Training	30 000			
- Consulting	80 000			
- Data Conversion	40 000			
Total	215 000			
Production Costs				
- Supplies		15 000	15 000	15 000
- IS Salaries		75 000	82 500	90 750
- Upgrades		10 000	10 000	10 000
Annual Prod.Costs (Present Value)		100 000 74 074	107 500 58 985	115 750 47 046
Accumulated Costs		289 074	348 059	395 105

PV = RM 47 046

(b)

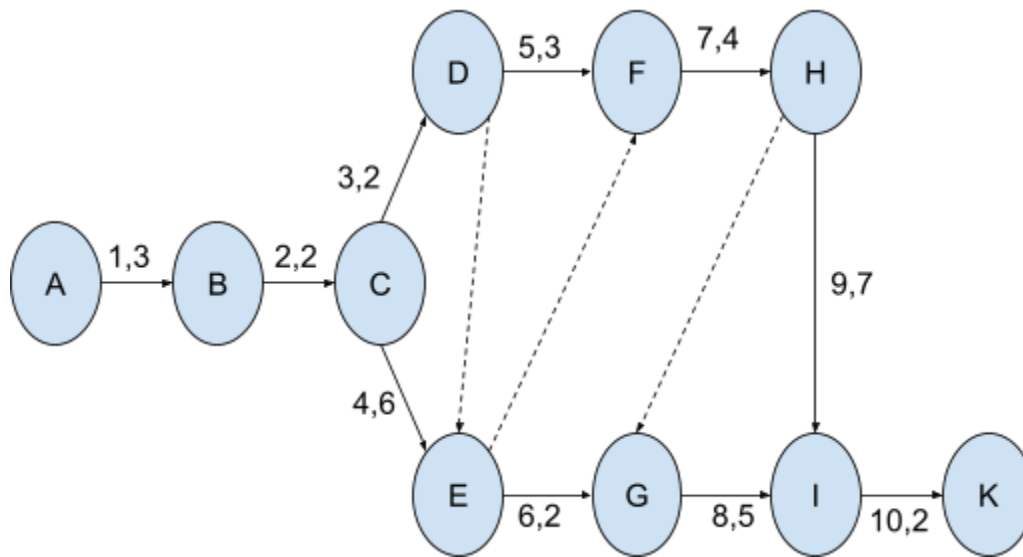
Benefits	Year 0	Year 1	Year 2	Year 3
- Improve Customer Service		150 000	180 000	216 000
- Increase productivity		150 000	187 500	234 375
Total		300 000	367 500	450 375
Present Value		222 222	201 646	183051
Accumulated benefits		222 222	423 868	606919
Gain or Loss		(66 852)	75 809	211 814

PI = 211 814 / 215 000
= 0.99

Profitability index = 0.99, showing that it is not very good investment because of its index is less than one.

Question 5

a.)



b.) Path 1= 1-2-3-5-7-9-10

Path 2= 1-2-3-5-7-8-10

Path 3= 1-2-3-6-8-10

Path 4= 1-2-4-7-9-10

Path 5= 1-2-4-7-8-10

Path 6= 1-2-4-6-8-10

c.) Path 1= $3+2+2+3+4+7+2= 23$ weeks

Path 2= $3+2+2+3+4+5+2= 21$ weeks

Path 3= $3+2+2+2+5+2= 16$ weeks

Path 4= $3+2+6+4+7+2= 24$ weeks

Path 5= $3+2+6+4+5+2= 22$ weeks

Path 6= $3+2+6+2+5+2= 20$ weeks

The critical path is path 4, 1-2-4-7-9-10, with a length of 24 weeks.

d.)

Immediate activity																								
Collect requirements																								
Analyze processes																								
Analyse data																								
Design process																								
Design data																								
Design screens																								
Design reports																								
Program																								
Test and document																								
Install																								
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
	Weeks																							

Weeks